

PRODUCT DOSSIER · ED-TECH

Everyone built an AI that teaches. Lucid built one that listens.

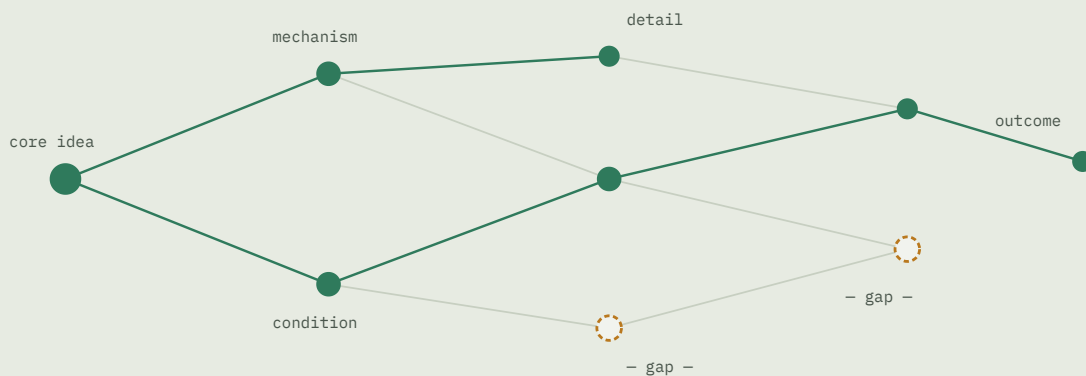
"The Tutor You Teach" — a learning tool built on the Feynman technique: you don't prove you understand something by reciting it back, you prove it by explaining it to someone confused, and having your explanation measured for what it's missing.

Next.js 16 · React 19

Claude / Gemini

transformers.js — MiniLM + DeBERTa NLI

Postgres (Neon)



coverage $\geq$ 0.55 cosine similarity · lit nodes are found in the student's own words

dashed = concept never mentioned → becomes the AI persona's next question

Recall is cheap. Understanding is expensive to fake out loud.

You can score full marks on a written test reciting a definition you can't actually use. Re-reading a page feels like learning, but the gaps only surface the moment you try to say the idea in your own words — which is exactly the moment most students never practice, until a viva or interview puts them on the spot.

i. Exams measure recall, not understanding Matching a definition to a blank is a different skill than being able to construct the idea from scratch.	ii. The illusion of competence Familiarity with a page feels like mastery. Explaining it out loud is where that illusion breaks — that's the point, not a flaw.
iii. Students never rehearse articulation A viva or interview is often the first time anyone has to say a concept aloud under pressure, with no script.	iv. Feedback loops are broken "Study harder" isn't actionable. "You never explained where the energy comes from" is.

A score the LLM never touches.

Lucid's central architectural bet: understanding is measured by a dedicated ML pipeline, not by asking an LLM to grade. The AI persona only ever asks about gaps the engine already found — it never invents the assessment.

Grasp Score — three measured dimensions, composed

Every explanation is scored 0–100 on how much of the concept it covers, how much of what it says is actually correct, and — once a short causal round runs — how deep the reasoning goes. A dimension that hasn't been measured yet is dropped from the formula and the rest reweight, rather than being scored as if it were perfect.

LIB/ML/SCORING.TS — COMPOSESCORE

```
grasp = round( (0.5·coverage + 0.3·correctness + 0.2·depth) / ac
```

Coverage		0.50
Correctness		0.30
Depth		0.20



Coverage — MiniLM-embeds every concept node and every clause of the answer; a node counts as covered above 0.55 cosine similarity to its closest clause.

Correctness — a DeBERTa NLI cross-encoder checks each clause for entailment/contradiction against the single closest reference fact — *only* the closest, to avoid the classic false-positive where NLI flags unrelated-but-true sentences as contradictions.

Parrot Detector — catching recitation, not just correctness

High similarity to the textbook alone doesn't mean copying — it might just mean the student was right. So the flag only fires when two independent signals agree:

semantic similarity to textbook
phrasing

> 0.72

AND

literal trigram overlap

> 0.28

"That's the textbook talking. Now you tell me — in your own words, with an example."

Removing the two "copying" features from the classifier — confirmed by an internal ablation study — collapses memorization-class recall from **0.97 to 0.34**, which is the empirical case for requiring both signals rather than either alone.

03 — HOW IT WORKS

Two people use Lucid: the one explaining, and the one reading the results.

SOLO STUDENT

1

Pick a concept

Choose one of six curated topics, or type any subject — an unrecognised topic is turned into a full concept map by the LLM on first request, then cached.

2

Explain it — typed or spoken

The explanation is segmented into clauses and scored purely by the ML engine. No LLM is involved in this step.

3

The AI persona asks one question

A confused-peer or examiner persona, grounded strictly in the detected gaps and wrong statements, asks a single two-sentence follow-up — never a new topic, never the underlying mechanism.

4

Loop Mode teaches the gaps, then re-tests

A short micro-lesson covers only what was missed; the student re-explains and the before/after Grasp Score delta becomes the feedback.

5

Report, then progress

The score gauge, dimension breakdown, and Parrot verdict land on a report page; history persists to the browser and resurfaces as a score-over-time chart with spaced-repetition scheduling.

TEACHER & CLASS

1

Assign a concept

Teacher picks a topic and mode; a six-character join code is generated from an alphabet that skips visually ambiguous characters — built to be read off a projector.

2

Students join and explain

Each student enters the same session experience under their name; results are written to Postgres instead of the browser, since a class's devices don't share local storage.

3

Teacher reads the pattern, not just the score

The dashboard aggregates the class average and — via the misconception taxonomy — which specific error patterns are recurring *across* the class, not just within one student.

04 — ARCHITECTURE

One deliberate rule: the LLM never scores anything.

Two LLM providers are supported behind one interface, chosen at startup by whichever API key is present. Gemini is preferred for its free tier and a comma-separated model fallback chain that survives per-model quota limits; Claude is the paid alternative. Either way, the LLM is used only for conversation, micro-lessons, and generating new concept maps — never for grading.

LAYER	CHOICE
Framework	Next.js 16.2 (App Router) · React 19.2 · TypeScript
LLM	Claude via @anthropic-ai/sdk · Gemini via raw REST
Understanding ML	@xenova/transformers — MiniLM-L6-v2 + NLI-DeBERTa-v3-xsmall, in-process
Database	pg → Postgres (Neon, pooled connection)
Validation	zod on every API input
Speech	Web Speech API (browser-native, Chromium only)
Offline training	Python — sentence-transformers, scikit-learn, pandas

Two persistence layers, deliberately different scopes

STORE	HOLDS	WHY
localStorage	Solo session history — score, dimensions, timestamps	No account system; single-browser is enough for one learner
Postgres	assignments, submissions	A teacher's dashboard and every student's device need to share one record

API surface

ROUTE	DOES
POST /api/analyze	Runs the ML scoring pipeline on an explanation — no LLM call
POST /api/concept	Resolves any topic string to a concept map, generating + caching one if new
POST /api/persona	Streams the AI persona's next question, grounded in detected gaps
POST /api/microlesson	Loop Mode's short remedial lesson, covering only the gaps
POST /api/assignment	Teacher creates an assignment; returns a join code

ROUTE	DOES
GET /api/assignment/[code]	Assignment + all submissions, for the join page and dashboard
POST /api/assignment/[code]/submit	Records one student's finished, scored session

05 — INSIDE THE SCORING ENGINE

Nine steps from raw sentences to a verdict.

`lib/ml/scoring.ts::runAnalysis` — the pipeline that runs on every submitted explanation.

1. **Segment** the explanation into clauses on strong boundaries only — a bare " and " isn't split, so "carbon dioxide and water" survives intact.
2. **Embed** every segment, concept node, textbook phrasing, misconception, and reference fact in parallel with MiniLM.
3. **Score coverage** — per node, the maximum cosine similarity across all segments; nodes above `0.55` count, weighted and summed.
4. **Compute distribution-shape features** — minimum, weakest-two-mean, median, the largest gap between consecutive similarities — because "misses 2–3 ideas" is a count, and one averaged coverage number washes that out.
5. **Run the Parrot Detector** — max semantic similarity to textbook phrasing, combined with trigram overlap, into a single two-signal flag.
6. **Check misconception similarity** — does any segment match a known wrong idea, independent of whether it also contradicts a fact.
7. **Run NLI correctness** — each segment checked only against its single closest reference fact, gated by a minimum similarity, specifically to dodge NLI's tendency to call unrelated true statements "contradictions."
8. **Classify** a 15-dimensional feature vector into `good` / `partial` / `memorized` / `wrong`, via a trained model with a transparent heuristic fallback if weights are missing.

9. **Score depth** (optional) — cosine similarity between two causal why-answers and their expected ideas, then compose the final Grasp Score.

06 – MISCONCEPTION TAXONOMY

Five ways a wrong answer is wrong.

Wrong statements aren't just flagged — they're classified into one of five subject-agnostic categories, which is what lets a teacher's dashboard show which error pattern is recurring across a whole class, not just within one student's answer.

term_conflation	Treating two distinct terms as identical — "atoms are the same thing as molecules."
overgeneralization	Applying an absolute rule outside its valid conditions — "all cells have a nucleus."
input_output_reversal	Getting the direction of a flow backwards — "plants breathe in oxygen and out CO ₂ during photosynthesis."
naive_causal_model	A mechanism that violates a conservation law — "a moving object eventually stops on its own, with no force."
definition_substitution	A wrong single-entity role — "the cytoplasm is empty space inside the cell."

07 – TRAINING THE CLASSIFIER

Trained offline in Python, deployed as a static weights file.

A synthetic dataset generator, a feature-extractor kept as a deliberate line-by-line mirror of the TypeScript scoring code, and a trainer that exports the winning model to `lib/ml/weights.json` for the app to load at runtime.

659	11	6	0.848
labeled samples	concepts	subjects	held-out accuracy

The dataset is LLM-authored and template-synthesized — no human-labeled test set exists yet, which the project's own documentation flags as an open item rather than glossing over.

08 — KNOWN LIMITATIONS

What the current build admits it doesn't do.

- ❗ Concept maps are the scoring ceiling — a poorly authored map caps how well any explanation can be assessed against it.
- ❗ One fixed coverage threshold (0.55) — a single unfine-tuned cosine cutoff is an admitted precision/recall tradeoff, not a calibrated boundary.
- ❗ English-only, tuned for roughly ages 14–18.
- ❗ Voice input needs Chromium — the Web Speech API has no Firefox implementation.
- ❗ Depth scoring is shallow — two similarity-scored questions, not a real reasoning assessment.

❗ No accounts — a teacher dashboard is gated only by knowledge of its join code; solo history lives in localStorage and is lost on cache clear.

❗ Classifier trained entirely on synthetic data — a human-labeled test set is a stated TODO, not yet built.

PREPARED FROM A DIRECT READ OF THE REPOSITORY — APP ROUTES, API HANDLERS, THE ML SCORING PIPELINE, AND THE TRAINING SCRIPTS — NOT FROM MARKETING COPY ALONE.